

Microbiome Composition Analyses of Obese *Drosophila* Populations Experiencing Heightened Early Age Mortality and Cardiac Dysfunction

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Obesity & Diabetes Implications on Health:

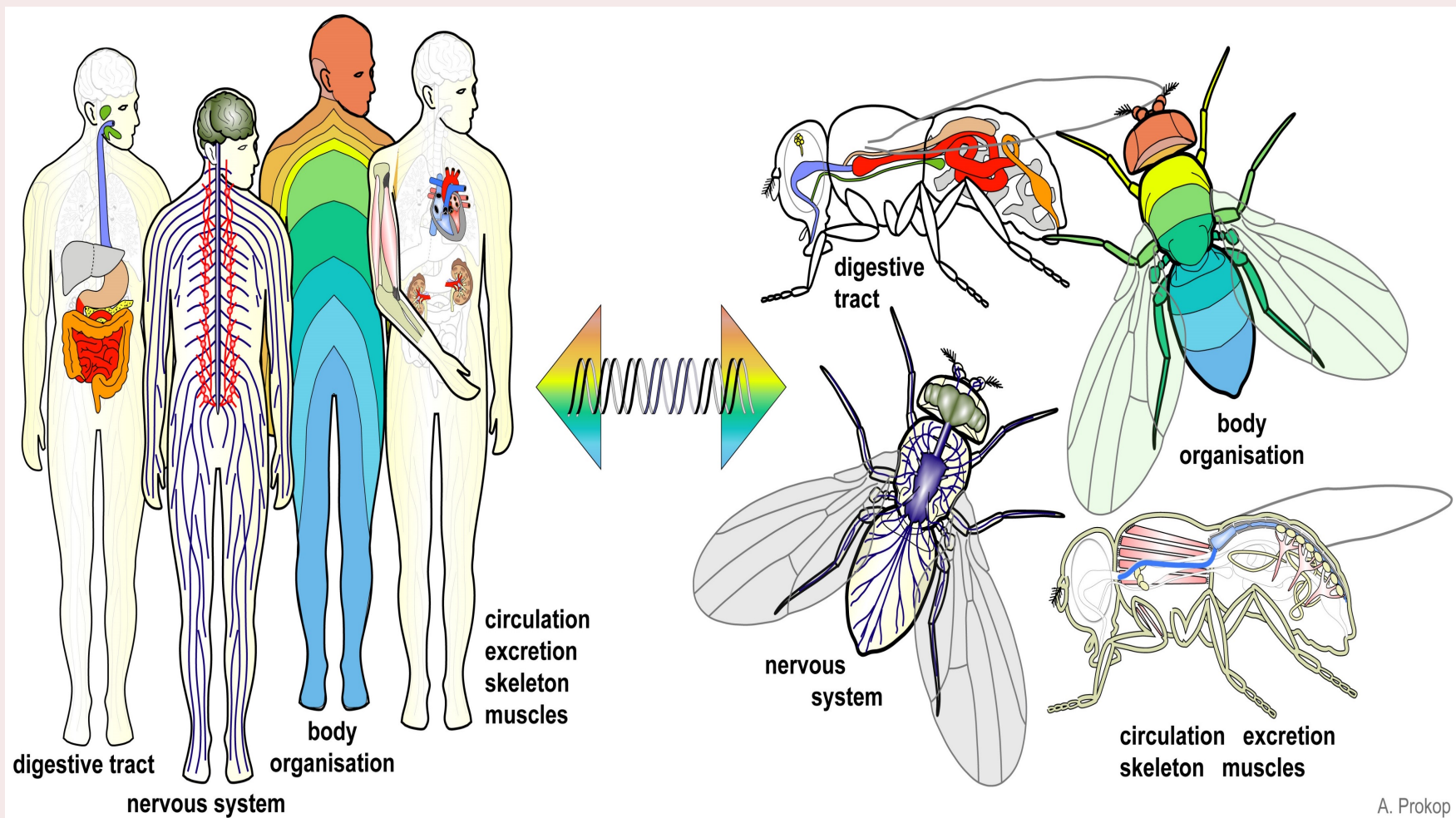
- Over recent decades, lifestyle changes have significantly increased the prevalence of obesity and diabetes in human populations.
- Two of the major causes of deaths in the U.S.
- Major risk factors for cardiovascular disease(s) and stroke.
- High-fat and high-sugar diets have been directly linked to these disorders.
- Occurrence may also be linked to genetics, race, age, sex, and socioeconomic status.
- Studying this in humans and vertebrate animals present major challenges, however, *Drosophila melanogaster* provides an opportunity to address these complex questions.
- Compared to humans, we also see similar cardiac dysfunction, weight gain, and shortened lifespans in either obese fruit fly populations or fruit flies fed a high-fat and high-sugar diet.
- Our lab focuses on identifying the key genetic and molecular predispositions that increases the rate(s) of developing these diseases.
- We utilize our 10 obese fruit fly populations and their respective control populations to model the obesity and diabetes phenotypes.

Key Questions and Goals:

Question: What differs in the gut microbial composition in our obese fly populations and control fly populations?

Long-term Goals: We want to connect the significant microbiome differences with genomic, transcriptomic, and proteomic signatures to better understand how these obesity and diabetes related phenotypes arise.

Advantages of *Drosophila* Research Model:



- Short generation time.
- Easily maintain hundreds of populations.
- Effective population size with abundant genetic variation.
- Observable evolutionary responses in as little as ~5-10 generations.
- High levels of conservation of gene/protein function between humans and flies.
- Range of diseases including, but not limited to, developmental, metabolic, cardiac, renal and neurodegenerative diseases.

Methodology:

Obesity Populations (S-type)

- In 2010, 10 control populations (C-type) were intensely selected for starvation resistance.
- Flies were exposed to nonnutritive agar until an 80% mortality threshold was reached.
- After 5 years of selection, mortality threshold increased from 3 days to 10 days in the obese SCO populations (S-type).

Table 1: Phenotypic characterization of the 50 experimentally evolved populations					
	A-type average (mean $\pm$ SE)	B-type average (mean $\pm$ SE)	C-type average (mean $\pm$ SE)	S-type average (mean $\pm$ SE)	T-type average (mean $\pm$ SE)
Cardiac arrest rate (%)	55.25 $\pm$ 1.766	37.5 $\pm$ 1.443	26.33 $\pm$ 3.5	53.33 $\pm$ 2.9	26.7 $\pm$.721
Survival during starvation (h)	34.142 $\pm$ 2.307	46.63 $\pm$ 2.03	64.14 $\pm$ 1.23	151.67 $\pm$ 4.23	71.293 $\pm$ 1.543
Survival during desiccation (h)	10.383 $\pm$.228	10.785 $\pm$.248	14.67 $\pm$.24	16.82 $\pm$.3	14.15 $\pm$.621
Lipid content (mg/fly)	.02 $\pm$.008	.032 $\pm$.002	.055 $\pm$.003	.141 $\pm$.006	...
Glycogen content (mg/fly)	.04 $\pm$.002	.053 $\pm$.004	.072 $\pm$.005	.157 $\pm$.009	...
Water content (mg/fly)	.586 $\pm$.01	.761 $\pm$.022	.891 $\pm$.01	1.076 $\pm$.012	...
Body mass (mg/fly)	.796 $\pm$.014	1.045 $\pm$.027	1.263 $\pm$.032	1.643 $\pm$.035	...

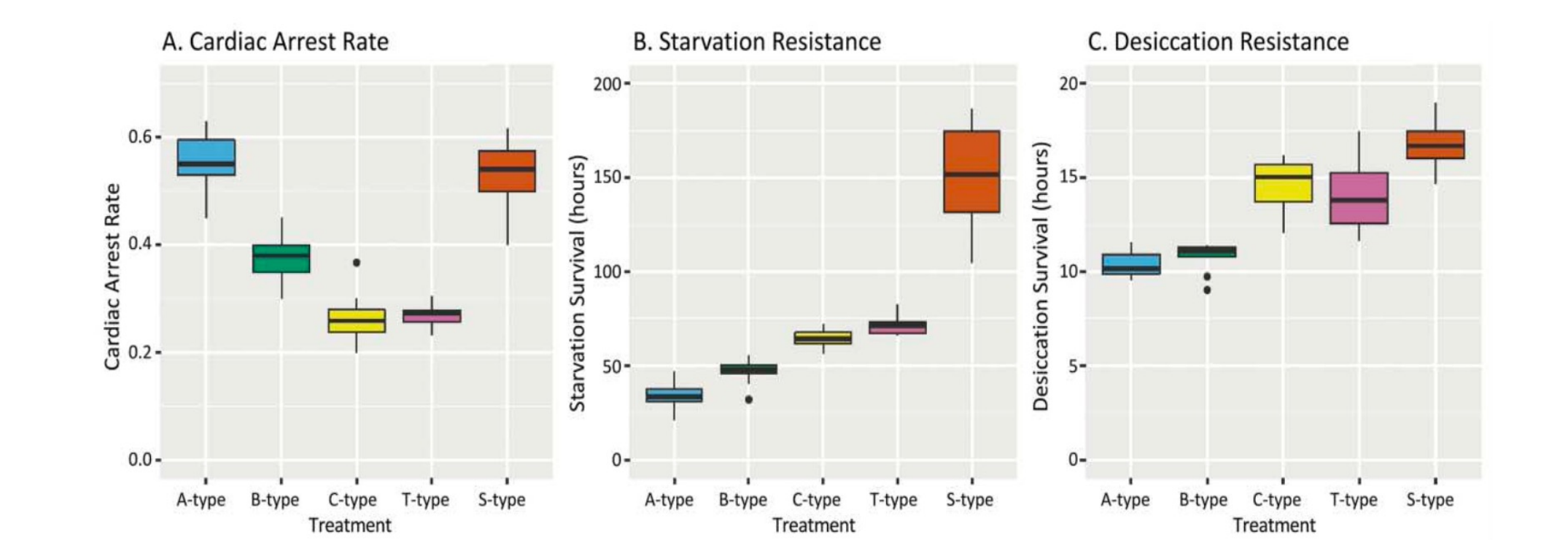


Figure 1. Table and Graphs of the Phenotypic Characteristics of the Obese Population
After five years of selection, the obese populations' (S-type) mortality threshold increased from three days to ten days. In addition to mortality difference, they also display a tripled lipid content in comparison to the C-type and a doubled cardiac arrest rate.

Experimental Approach:

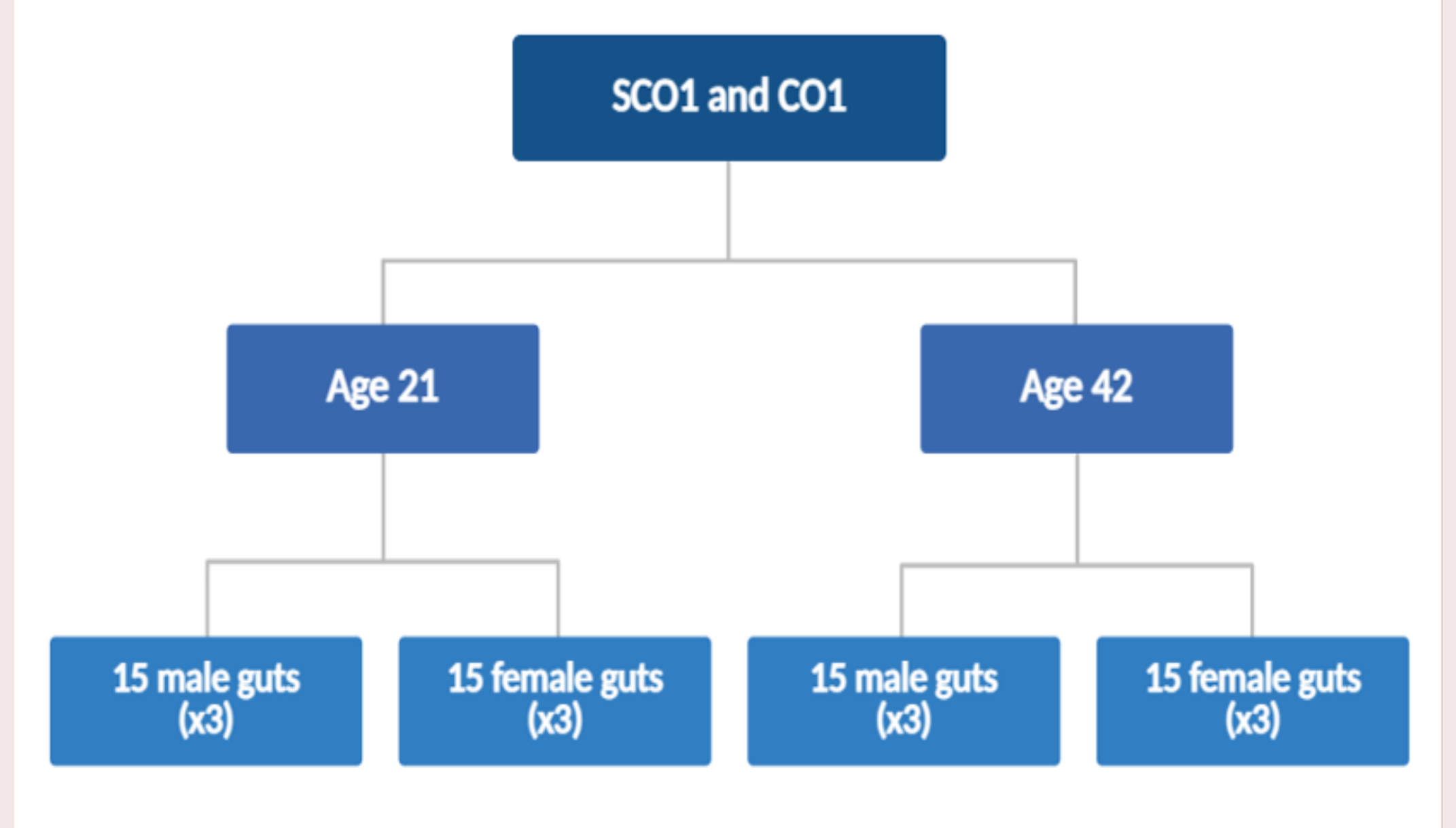
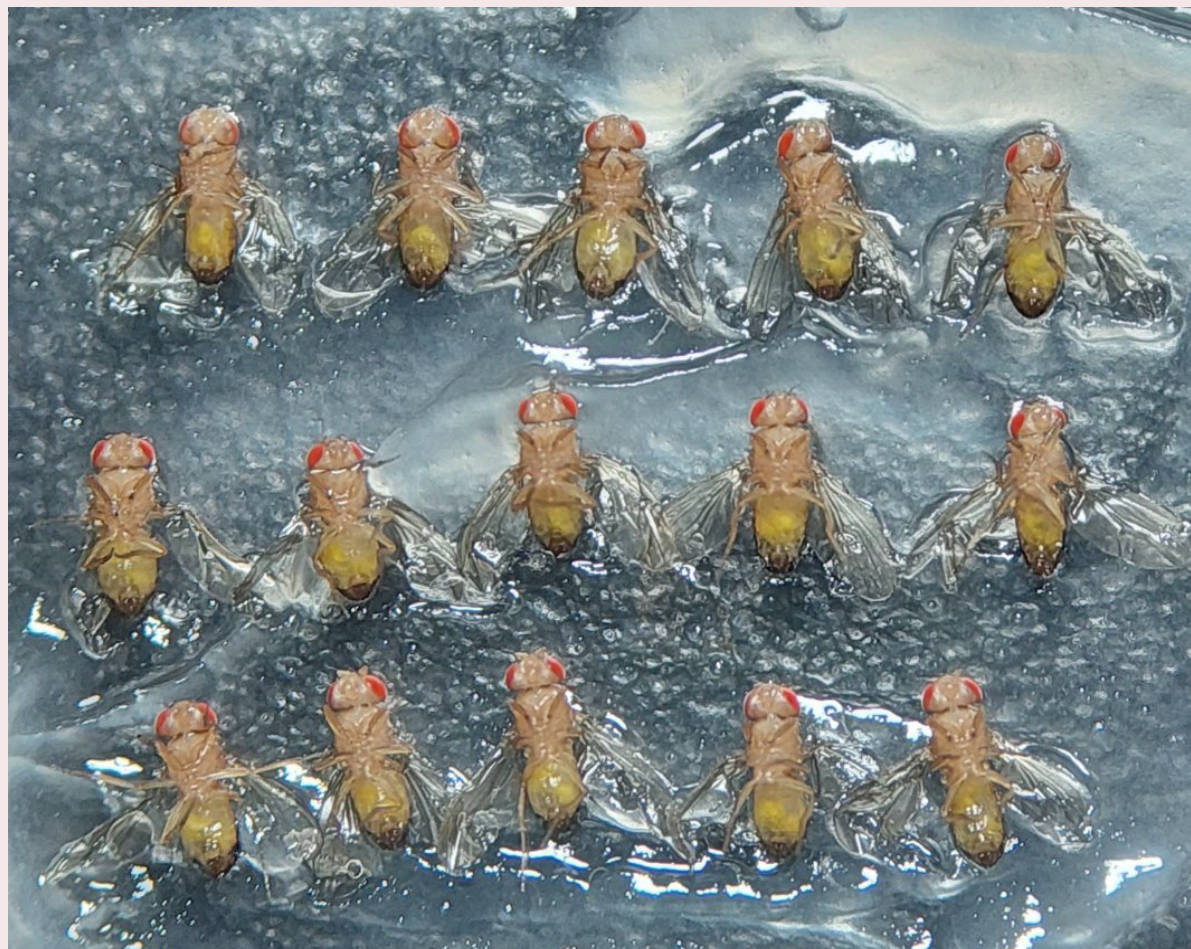


Figure 2. An Example Flow Chart Depicting the Gut Collection for Each Population
Two populations (SCO1 & CO1) are depicted with each age point for gut collection. At age Day 21 and Day 42, 15 male guts will be collected (x3) and 15 female guts will be collected (x3). As of today, 2,000 guts have been pulled, and there will be a total of 3,600 guts pulled.

Fly Dissection and Gut Extraction:

Step 1. Mount flies in a tissue culture dish prepared with petroleum jelly.



Step 2. Remove head and top of thorax, including all legs.



Step 3. Open abdomen to expose the intestines, Malpighian tubules, and body fat.



Step 4. Extract intestines and Malpighian tubules, leaving behind all bodily fat to minimize host DNA contamination.



Determining Gut Quantity for Sufficient DNA Concentration:

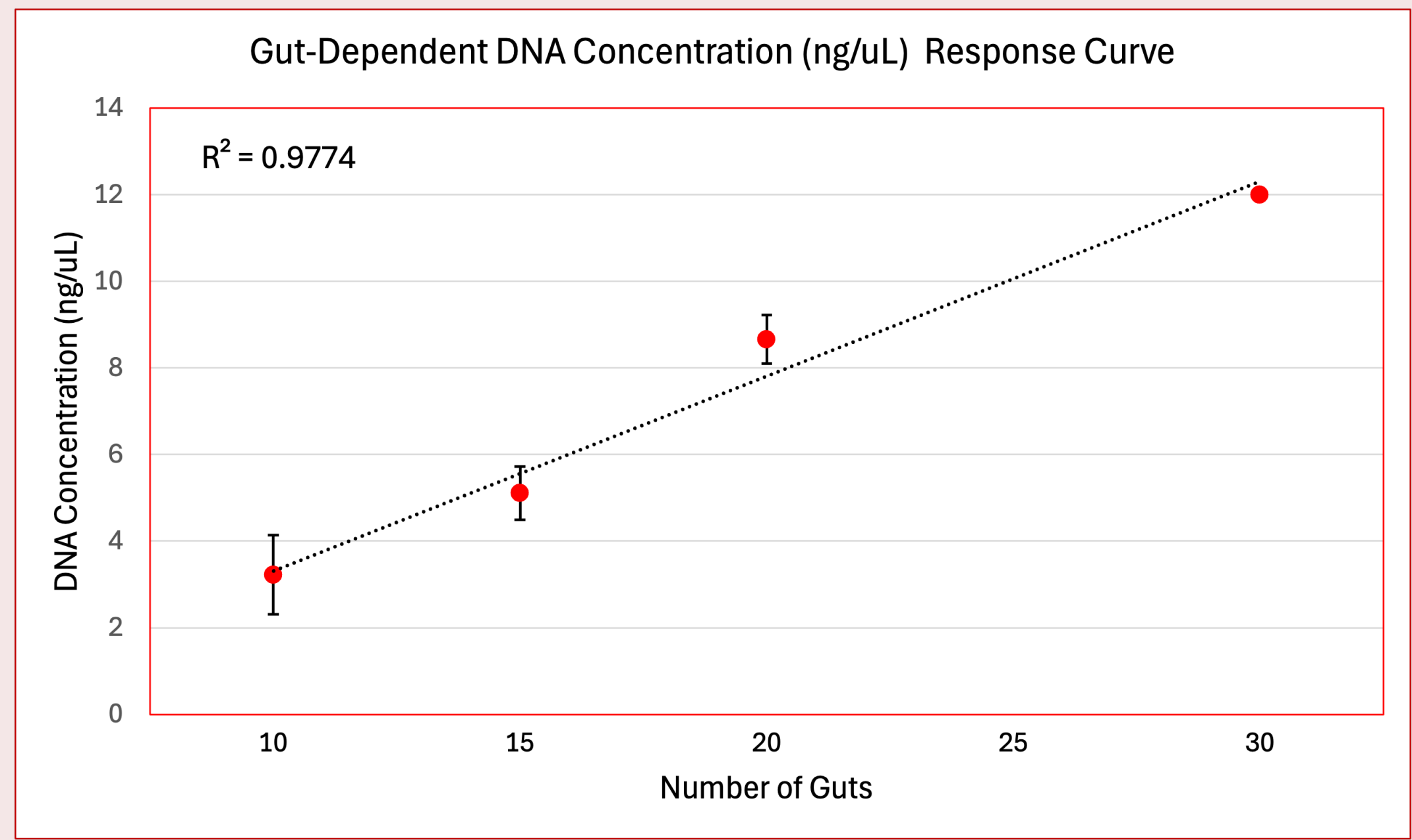
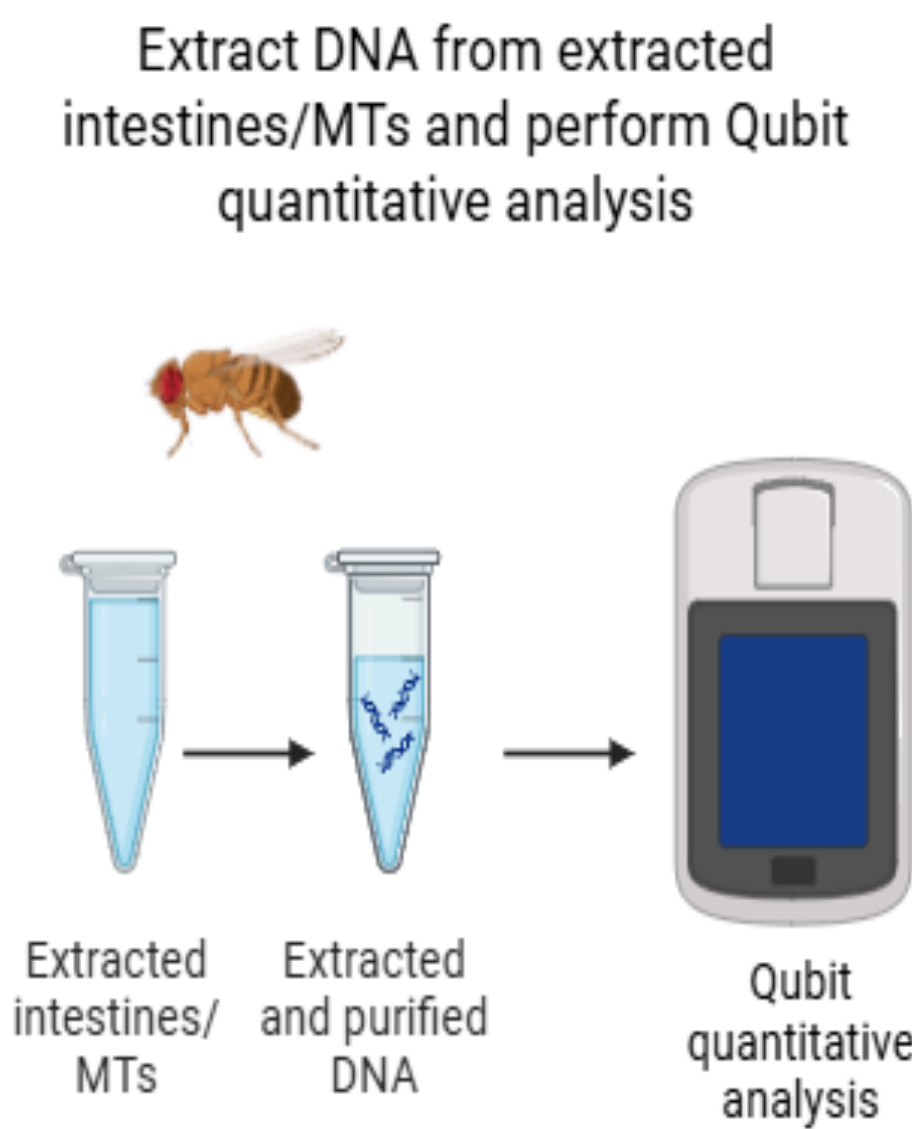


Figure 3. Gut-Dependent DNA Concentration (ng/uL) Response Curve
The concentration of DNA varies naturally per sample, so the exact quantity of guts needed was unknown. We performed a trial DNA extraction with samples of various amounts of guts. We found that 15 guts produced adequate DNA for sequencing pipeline.

DNA Extraction and Sequencing Pipeline:

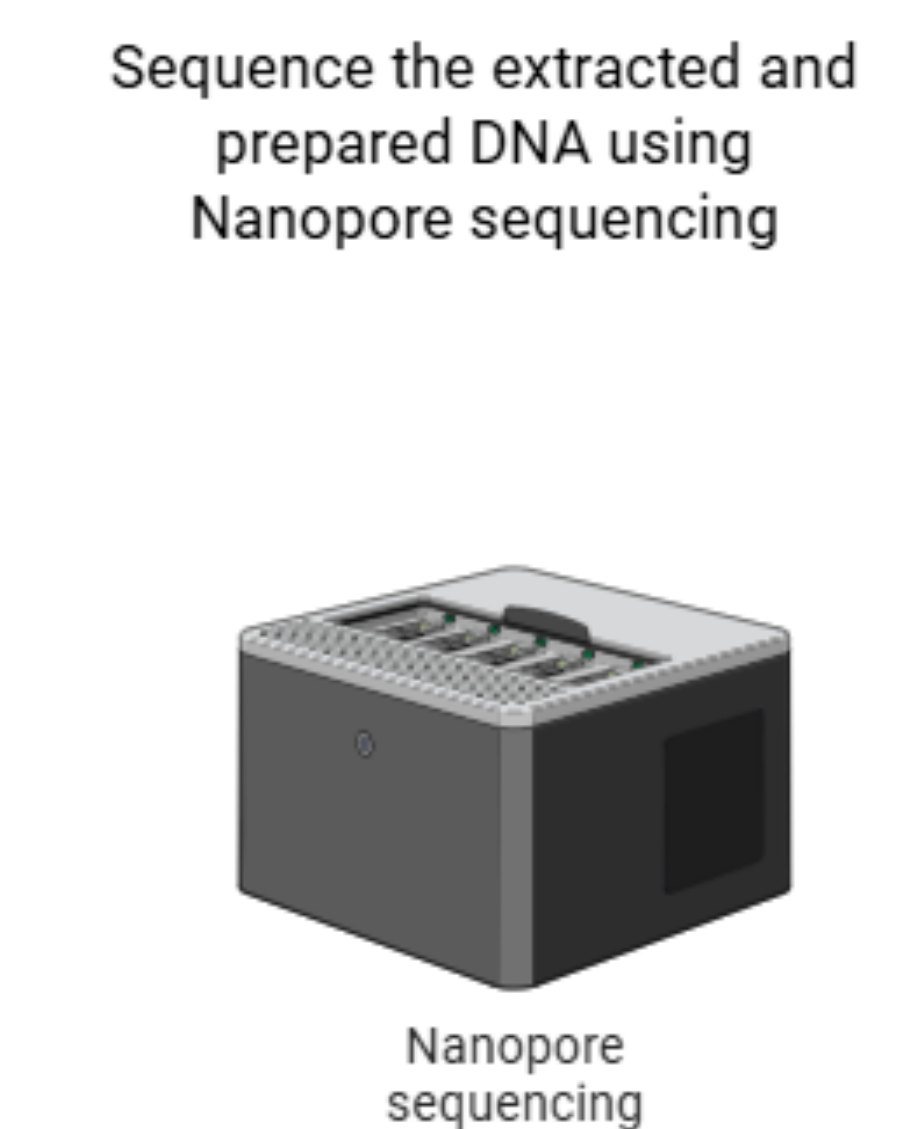
Step 1. Extract DNA from the extracted guts and perform Qubit quantitative analysis on the purified DNA sample.



Step 2. Perform library preparation on the purified DNA and attach adapters for further sequencing.



Step 3. Sequence the prepared DNA using Nanopore sequencing, also known as 16s rRNA sequencing.



Step 4. Perform taxonomical alignment and analyses via R.



Expected Outcomes and Future Directions:

- Age 21 flies are expected to show large differences in microbial diversity between groups, which are predicted to diminish or disappear with age.
- Follow-up shotgun sequencing is planned to achieve deeper taxonomic resolution beyond 16s rRNA data.
- Microbiome data will be merged with other multi-omic/biological layers to better understand obesity-related phenotypes.

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